

Nonlinear Dynamics & Chaos HW4

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Technion

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1 The (well named) Brusselator

1.1 Fixed Points and Stability

at fixed points

$$\dot{x} = 0, \quad \dot{y} = 0 \quad (1)$$

so

$$\dot{x} + \dot{y} = 0 \quad (2)$$

inserting the equations

$$a - x = 0 \implies x^* = a \quad (3)$$

inserting into the equation for y

$$a(b - ay^*) = 0 \implies y^* = \frac{b}{a} \quad (4)$$

giving the fixed point

$$(x^*, y^*) = \left(a, \frac{b}{a}\right) \quad (5)$$

onto stability. The jacobian is

$$J = \begin{pmatrix} -(b+1) + 2xy & x^2 \\ b - 2xy & -x^2 \end{pmatrix} \quad (6)$$

and at the fixed point

$$J_* = \begin{pmatrix} b-1 & a^2 \\ -b & -a^2 \end{pmatrix} \quad (7)$$

with trace and determinant

$$t = \text{tr } J_* = b - b - 1 - a^2, \quad d = \det J_* = a^2 \quad (8)$$

so the eigenvalues are

$$\lambda_{\pm} = \frac{t \pm \sqrt{t^2 - 4a^2}}{2} \quad (9)$$

since $d > 0$ the stability is controlled by the trace

$$\begin{cases} \text{stable fixed point} & b < 1 + a^2 \\ \text{unstable fixed point} & b > 1 + a^2 \end{cases} \quad (10)$$

1.2 Bifurcation

A Hopf bifurcation happens when the trace vanishes while the determinant stays positive so the critical b is

$$b_c - 1 - a^2 = 0 \implies b_c = 1 + a^2 \quad (11)$$

at which point

$$\lambda_{\pm} = \pm ia \quad (12)$$

so the frequency is

$$\omega_0 = a \quad (13)$$

1.3 Which side has the cycle?

The fixed point we found is unstable for $b > b_c$, and it's contained in a bounded trapping region. Any trajectory inside the region can't escape, but it also can't converge to the fixed point since it's unstable. Since there are no other fixed point, the poincare-bendixson theorem implies that it must approach a closed orbit. This means a stable limit cycle exists for $b > b_c$. In turn, this means the cycle is born on the side where the fixed point becomes unstable, so it's a supercritical hpof bifurcation.

1.4 Numerics

1.4.1 plot

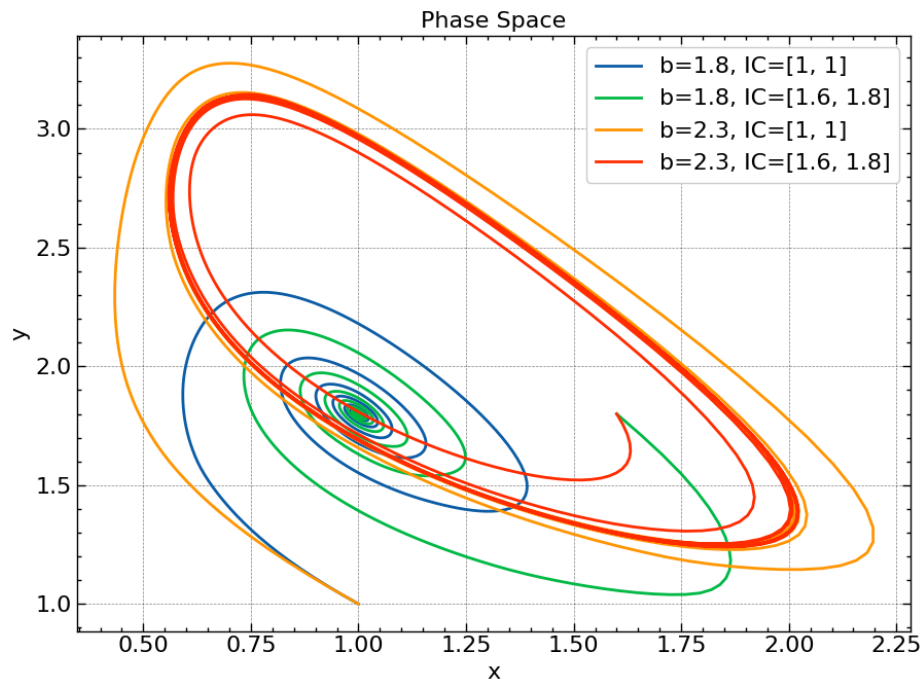


Figure 1 – Phase space drawing of the brusselator with different values of b and initial conditions. The behavior is indeed consistent with what I expected.

1.4.2 amplitude

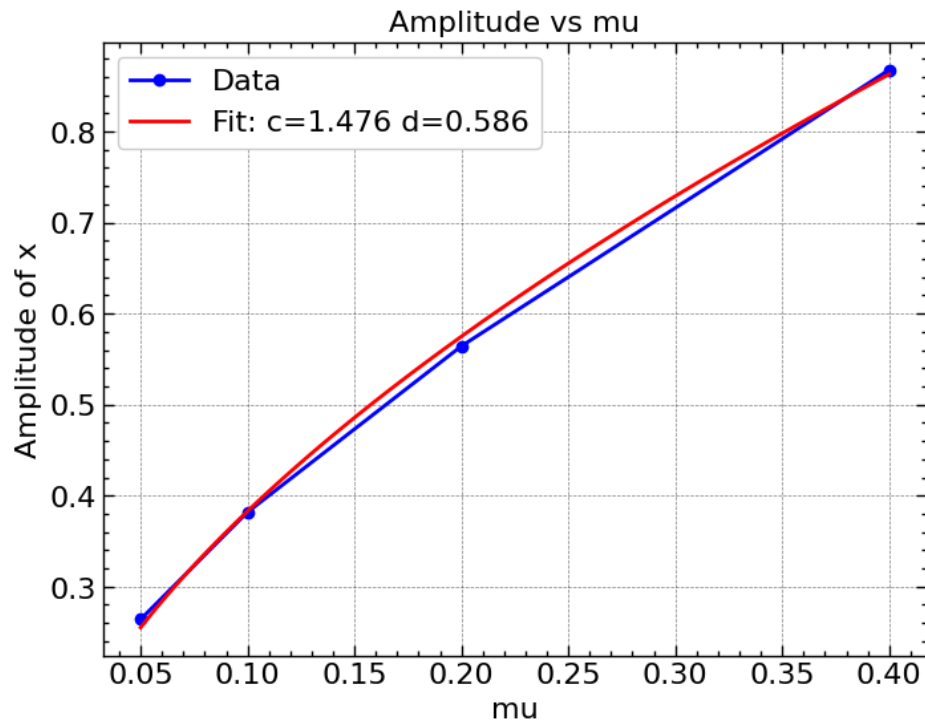


Figure 2 – Amplitude as a function of the value of μ . The function $y = c \cdot \mu^d$ is fitted. The resulting values aren't the exact theoretical value but are close enough. After all, we have 4 points to go off of.

1.4.3 Period

The period calculated for $\mu = 0.05$ is 6.3. The expectation was $T = 2\pi/a = 2\pi = 6.28$ so it's again pretty good.

2 Infinite-period bifurcations

2.1 saddle node on a cycle

2.1.1 local normal form

Just after the saddle node, the two fixed point disappear. After rescaling u , time, and μ the normal form is

$$\dot{u} = \mu + u^2 \quad (14)$$

2.1.2 time in the bottleneck

The time to move from $-h$ to h is

$$T = \int_{-h}^h \frac{du}{\mu + u^2} \quad (15)$$

using the known (to google and now to me) rule for the antiderivative of something of this form

$$T = \left(\frac{1}{\sqrt{\mu}} \arctan \left(\frac{u}{\sqrt{\mu}} \right) \right) \Big|_{-h}^h = \frac{2}{\sqrt{\mu}} \arctan \left(\frac{h}{\sqrt{\mu}} \right) \quad (16)$$

so in the limit $\mu \rightarrow 0^+$

$$T \sim \frac{\pi}{\sqrt{\mu}} \quad (17)$$

2.2 Homoclinic saddle-loop

2.2.1 entry point consistent

at the bifurcation

$$\mu = \mu_c \implies \varepsilon = 0 \quad (18)$$

the trajectory enters exactly on the stable manifold $u = 0$, and approaches the saddle asymptotically, leaving along the unstable manifold.

2.2.2 escape time

The unstable coordinate evolves

$$u(t) = \varepsilon e^{\lambda_u t} \quad (19)$$

and the trajectory leaves when $u(t) = h$ i.e

$$h = \varepsilon e^{\lambda_u T} \quad (20)$$

so

$$T = \frac{1}{\lambda_u} \ln \left(\frac{h}{\varepsilon} \right) \quad (21)$$

inserting $\varepsilon = u_0 |\mu - \mu_c|$

$$T = \frac{1}{\lambda_u} \left(\ln \left(\frac{h}{u_0} \right) + \ln \left(\frac{1}{|\mu - \mu_c|} \right) \right) \quad (22)$$

the first term is a constant so

$$T \sim \frac{1}{\lambda_u} \ln \left(\frac{1}{|\mu - \mu_c|} \right) \quad (23)$$

so

$$C = \frac{1}{\lambda_u} \quad (24)$$

2.2.3 smallest distance

the stable coordinate is

$$w(t) = h e^{\lambda_s t} \quad (25)$$

since $\lambda_s < 0$, w decreases while the orbit remains in the neighborhood. The smallest value occurs at the escape time

$$w_{\min} = h e^{\lambda_s T} \quad (26)$$

inserting the time we found

$$w_{\min} = h \left(\frac{h}{\varepsilon} \right)^{\lambda_s / \lambda_u} \quad (27)$$

so

$$w_{\min} = h \left(\frac{h}{u_0 |\mu - \mu_c|} \right)^{\lambda_s / \lambda_u} \quad (28)$$

2.3 origin of the diverging periods

In the SNIC, the local velocity vanishes quadratically at the saddle node. In the homoclinic bifurcation, the orbit enters exponentially close to the stable manifold, therefore spending a very long time near the saddle.

3 Synchronization from averaging for a weakly forced oscillator

3.1 Main synchronization region

3.1.1 slow phase

Since we define

$$\psi = \theta - \Omega t \quad (29)$$

then

$$\dot{\psi} = \dot{\theta} - \Omega = \Delta\omega + \varepsilon Q(\psi + \Omega t, \Omega t) \quad (30)$$

both of these terms are $O(\varepsilon)$ so

$$\dot{\psi} = O(\varepsilon) \quad (31)$$

so over one period

$$\Delta\psi = O(\varepsilon T) \quad (32)$$

meaning ψ is constant to leading order during a single period.

3.1.2 Averaged equation

inserting theta and phi

$$Q(\psi + \Omega t, \Omega t) = \sum_{m,n} c_{m,n} e^{im\psi} e^{i(m+n)\Omega t} \quad (33)$$

averaging over one period means that only the terms for which $m + n = 0$ survive which eliminates the dependence on Ω , meaning

$$\dot{\psi} = \Delta\omega + \varepsilon f(\psi) \quad (34)$$

where

$$f(\psi) = \sum_m c_{m,-m} e^{im\psi} \quad (35)$$

3.1.3 locking range

Since a locked solution is a fixed point of the averaged relative phase equation

$$0 = \Delta\omega + \varepsilon f(\psi^*) \quad (36)$$

locking happens when

$$f_{\min} \leq -\frac{\Delta\omega}{\varepsilon} \leq f_{\max} \quad (37)$$

so

$$-\varepsilon f_{\max} \leq \Delta\omega \leq -\varepsilon f_{\min} \quad (38)$$

so the locking width is

$$\Delta\omega_{\text{width}} = \varepsilon (f_{\max} - f_{\min}) \quad (39)$$

and the fixed point is stable when

$$\varepsilon f'(\psi^*) < 0 \quad (40)$$

3.2 p:q resonances

defining

$$\psi = q\theta - p\Omega t \quad (41)$$

and assuming

$$q\omega_0 \approx p\Omega \quad (42)$$

means

$$\dot{\psi} = q\dot{\theta} - p\Omega = \delta_{pq} + \varepsilon q Q(\theta, \Omega t) \quad (43)$$

if we define $\phi = \Omega t$ then

$$\theta = \frac{\psi + p\phi}{q} \quad (44)$$

which means

$$\dot{\psi} = \delta_{pq} + \varepsilon q \sum_{m,n} c_{m,n} e^{im\psi/q} e^{i(mp/q+n)\phi} \quad (45)$$

and again when averaging the only terms that survive are those where

$$\frac{mp}{q} + n = 0 \implies mp + nq = 0 \quad (46)$$

but since p and q are coprime

$$m = kq, \quad n = -kp \quad (47)$$

so

$$\dot{\psi} = \delta_{pq} + \varepsilon f_{pq}(\psi) \quad (48)$$

where

$$f_{pq}(\psi) = q \sum_k c_{kq, -kp} e^{ik\psi} \quad (49)$$

3.3 The Adler equation

The Adler model is

$$\dot{\theta} = \omega_0 - \varepsilon \sin(\theta - \Omega t) \quad (50)$$

so

$$Q(\theta, \phi) = -\sin(\theta - \phi) \quad (51)$$

which if we expand this in a fourier expansion, it'll contain only the harmonics

$$(m, n) = (1, 1), \quad (-1, 1) \quad (52)$$

but for a p:q resonance the coefficients need the form

$$(m, n) = (kq, -kp) \quad (53)$$

which are satisfied for the Adler harmonics only for

$$p = q = 1 \quad (54)$$

as a result, for every $(p, q) \neq (1, 1)$

$$f_{pq} = 0 \tag{55}$$

making the averaged equation

$$\dot{\psi} = \delta_{pq} \tag{56}$$

meaning there is no finite locking interval, meaning the adler equation has no finite-width higher p:q resonances.

4 Saddle-node bifurcations in the sine-circle map

4.1 1:1 locking

4.1.1 fixed points

The synchronization condition is

$$\theta^* + 2\pi = G(\theta^*) \quad (57)$$

taking modulo 2π it's exactly

$$P(\theta^*) = \theta^* \quad (58)$$

so θ^* is a fixed point of the circle map. Substituting G

$$\theta^* + 2\pi = \theta^* + \eta - K \sin \theta^* \quad (59)$$

so

$$\sin \theta^* = \frac{\eta - 2\pi}{K} \quad (60)$$

so solutions exist when

$$|\eta - 2\pi| \leq K \quad (61)$$

4.1.2 stability

the two solutions in the interval are

$$\theta_1 = \arcsin\left(\frac{\eta - 2\pi}{K}\right), \quad \theta_2 = \pi - \arcsin\left(\frac{\eta - 2\pi}{K}\right) \quad (62)$$

the multiplier is

$$\Lambda = G'(\theta^*) = 1 - K \cos \theta^* \quad (63)$$

since θ_1 is positive, $\cos \theta_1$ is positive, and since $0 < K < 1$

$$0 < 1 - K \cos \theta_1 < 1 \quad (64)$$

so θ_1 is stable. With similar arguments θ_2 is unstable.

4.1.3 Tanget bifurcation

At the upper edge

$$\eta = 2\pi + K \quad (65)$$

so

$$\sin \theta^* = 1 \implies \theta^* = \pi/2 \quad (66)$$

and at the lower edge

$$\eta = 2\pi - K \quad (67)$$

so

$$\sin \theta^* = -1 \implies \theta^* = 3\pi/2 \quad (68)$$

but

$$G'(\theta^*) = 1 - K \cos \theta^* = 1 \quad (69)$$

and

$$G''(\theta^*) = K \sin \theta^* \neq 0 \quad (70)$$

so G is tangent to the line $\theta + 2\pi$, which confirms it's a saddle node bifurcation.

4.2 1:2 tongue

4.2.1 second iterate

The first iterate is

$$\theta_1 = \theta_0 + \pi + \delta - K \sin \theta_0 \quad (71)$$

the second iterate is

$$\theta_2 = \theta_1 + \pi + \delta - K \sin \theta_1 = \theta_0 + 2\pi + 2\delta - K \sin \theta_0 - K \sin (\theta_0 + \pi + \delta - K \sin \theta_0) \quad (72)$$

using $\sin(x + \pi) = -\sin x$

$$\theta_2 = \theta_0 + 2\pi + 2\delta - K \sin \theta_0 + K \sin (\theta_0 + \delta - K \sin \theta_0) \quad (73)$$

expanding the sine

$$\sin(\theta + \delta - K \sin \theta) \simeq \sin \theta + (\delta - K \sin \theta) \cos \theta \quad (74)$$

so

$$\theta_2 \simeq \theta_0 + 2\pi + 2\delta_K \delta \cos \theta_0 - K^2 \sin \theta_0 \cos \theta_0 \quad (75)$$

but since δ will be $O(K^2)$ so in leading order

$$G^{(2)}(\theta) \simeq \theta + 2\pi + 2\delta - \frac{K^2}{2} \sin 2\theta \quad (76)$$

4.2.2 synchronization interval

the synchronization condition is

$$G^{(2)}(\theta^*) = \theta^* + 2\pi \quad (77)$$

so

$$2\delta - \frac{K^2}{2} \sin 2\theta^* = 0 \implies \sin 2\theta^* = \frac{4\delta}{K^2} \quad (78)$$

solutions exist when

$$\left| \frac{4\delta}{K^2} \right| \leq 1 \quad (79)$$

so

$$|\delta| \leq \frac{K^2}{4} \quad (80)$$

so the width is

$$2|\delta| = \frac{K^2}{2} \quad (81)$$

so it scales with the square of K , compared to it scaling with K for 1:1.

4.2.3 Saddle-node points

at a saddle node

$$G^{(2)}(\theta) - \theta - 2\pi \simeq 2\delta - \frac{K^2}{2} \sin 2\theta = 0 \quad (82)$$

and

$$\frac{\partial}{\partial \theta} \left(2\delta - \frac{K^2}{2} \sin 2\theta \right) = -K^2 \cos 2\theta = 0 \quad (83)$$

so

$$\cos 2\theta = 0 \implies \theta = \pi/4 + n\pi/2 \quad (84)$$

at the upper edge

$$\delta = \frac{K^2}{4}, \quad \sin 2\theta = 1 \quad (85)$$

so

$$\theta_0 = \pi/4, \quad 5\pi/4 \quad (86)$$

and at the lower edge

$$\delta = -\frac{K^2}{4}, \quad \sin 2\theta = -1 \quad (87)$$

so

$$\theta_0 = 3\pi/4, \quad 7\pi/4 \quad (88)$$

both the points at each boundary are members of the same period-two orbit, at which $(G^{(2)})'(\theta) = 1$ and the second derivative is zero, confirming saddle-node bifurcations.

4.2.4 exact second-iterate plot

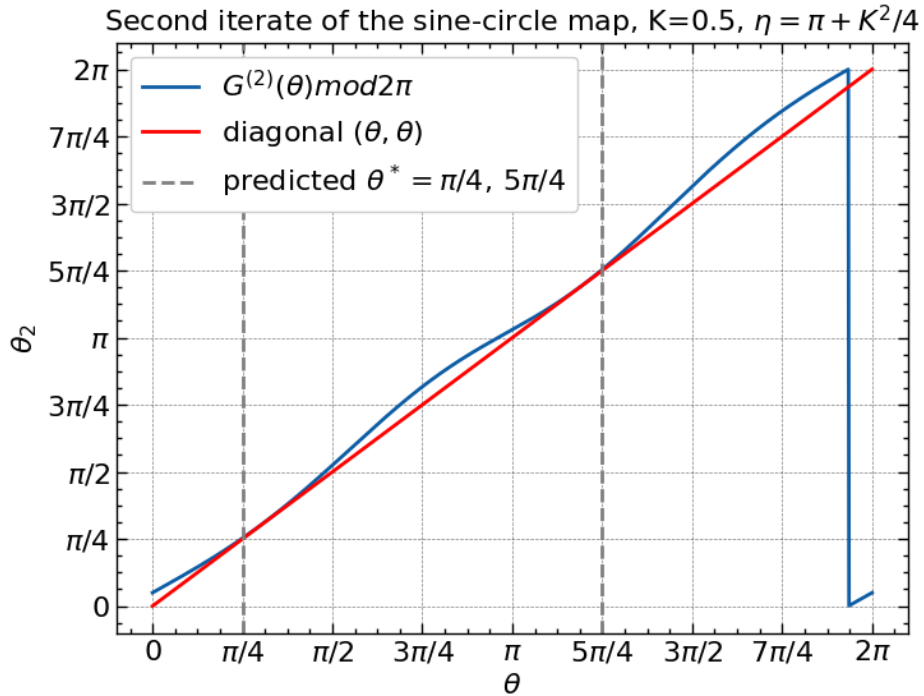


Figure 3 – Exact second iterate of the sine-circle map for $K = 0.5$.

4.2.5 Replacing $\sin \theta$ with $\sin 2\theta$

The first iterate is

$$\theta_1 = \theta + \pi + \delta + K \sin 2\theta \quad (89)$$

and the second

$$\theta_2 = \theta + 2\pi + 2\delta + K \sin 2\theta + K \sin (2\theta + 2\pi + 2\delta + 2K \sin 2\theta) \quad (90)$$

but to leading first order

$$\sin (2\theta + 2\pi + 2\delta + 2K \sin 2\theta) \simeq \sin 2\theta \quad (91)$$

so

$$\theta_2 \simeq \theta + 2\pi + 2\delta + 2K \sin 2\theta \quad (92)$$

and the locking condition becomes

$$2\delta + 2K \sin 2\theta^* = 0 \quad (93)$$

or

$$\sin 2\theta^* = -\delta/K \quad (94)$$

so

$$|\delta| \leq |K| \quad (95)$$

giving the full width

$$2|\delta| = 2|K| \quad (96)$$

so it goes like K this time. The difference comes from the behavior under half a turn. When the angle is multiplied by 2, the half turns instead of contributing a negative contribution, they add up positively.